UNIVERSITY OF KARACHI

Department of Computer Science (UBIT)

LAB NO. 01

Operating Systems (SE-458)

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LAB NO. 01 - INTRODUCTION TO OPERATING SYSTEMS

Objective

To study the history, functions and types of operating systems, and compare DOS, Windows and Linux.

Question 1: Briefly describe the history of operating systems.

An operating system is system software that manages computer hardware and provides services to programs. Operating systems developed as computers needed to handle more work and support more users.

Early Computers

In the 1940s and early 1950s, many computers had no operating system. Operators loaded programs and controlled their execution manually.

Batch Processing and Time-Sharing

During the 1950s, batch systems began processing groups of jobs with less manual work between them. In the 1960s, multiprogramming allowed several programs to remain in memory, while time-sharing allowed users to interact with a computer by sharing processor time.

UNIX and Personal Computers

UNIX development began at Bell Labs in 1969, with Ken Thompson and Dennis Ritchie playing major roles. It became known for supporting multiple users and tasks.

As personal computers spread, systems such as DOS and Windows made computers available for individual use. DOS mainly used typed commands, while Windows provided a graphical interface.

Development of Linux

Richard Stallman announced the GNU Project in 1983 to develop a free Unix-like operating system. Andrew Tanenbaum released MINIX in 1987 for teaching operating-system design.

Linus Torvalds began developing the Linux kernel in 1991, and Linux version 1.0 was released in 1994. The kernel was combined with GNU tools and other software to form complete distributions, such as Debian and Ubuntu. Linux is used in servers, desktops, embedded devices and Android systems.

Question 2: Explain the main functions of an operating system.

The operating system manages hardware resources and provides a working environment for applications. Its main functions are:

1. Processor Management

Schedules processes and assigns CPU time so different programs can make progress.

2. Memory Management

Allocates memory to processes, tracks its use and protects one process's memory from unauthorized access by another.

3. Input/Output Management

Coordinates communication with devices such as keyboards, displays and printers through device drivers.

4. Application Execution

Loads and starts programs, manages their resources and releases those resources when the programs finish.

5. Authorization and Protection

Uses accounts and permissions to control access to files, programs and other resources.

6. File Management

Organizes files and directories and supports creating, reading, writing, copying, renaming and deleting files.

7. System Information and Error Handling

Provides information about system activity, detects errors and reports problems such as failed device operations.

For example, when a browser and a word processor run together, the OS shares processor time between them, manages their memory and handles their input/output requests.

Question 3: Describe the different types and capabilities of operating systems.

Operating systems can be classified by their interface, users and execution capabilities. A single operating system may support several of the following features.

1. GUI-Based Operating System

A graphical user interface allows users to work through windows, icons, menus and buttons.

Example: Windows or Ubuntu with a graphical desktop.

2. Multi-User Operating System

A multi-user system allows several users to access the same computer through separate accounts. It manages their resources and access permissions.

Example: Several students logged into a Linux server.

3. Multitasking Operating System

A multitasking system allows several tasks to make progress during the same period. On one CPU core, the OS switches between tasks.

Example: Editing a document while downloading a file.

4. Multiprocessing Operating System

A multiprocessing system supports multiple processors or CPU cores, allowing work to execute in parallel.

Example: A program using several CPU cores to process a video.

5. Multithreading Operating System

A multithreading system supports multiple threads within a process. Different threads can handle different parts of a program's work.

Example: One thread handles user input while another performs a background calculation.

GUI describes the interface, while the other terms describe system capabilities. Windows and Linux can provide a GUI while also supporting multitasking, multiprocessing and multithreading. Classic MS-DOS, in comparison, is mainly a single-user, single-tasking system.

Question 4: Differentiate between DOS, Windows and Linux.

DOS stands for Disk Operating System. The comparison below refers to classic MS-DOS, modern Windows and common Linux distributions such as Ubuntu.

Linux is technically the kernel; a distribution combines it with tools and applications to provide a complete operating system. Early Windows versions worked with DOS, but modern Windows does not depend on MS-DOS.

1. User Interface

• MS-DOS: Command-line interface (CLI) only, where commands must be manually typed.

• Windows: Graphical User Interface (GUI) with icons, windows, and mouse navigation, along with command-line tools (Command Prompt, PowerShell).

• Linux: Offers both powerful Command-Line Interfaces (Bash/Shell) and customizable Graphical User Interfaces (GNOME, KDE, XFCE).

2. Users and Multitasking

• MS-DOS: Single-user, single-tasking system (executes one program at a time).

• Windows: Multi-user and preemptive multitasking system.

• Linux: True multi-user, multi-tasking, and multi-processing operating system built for concurrent execution.

3. Source Code and Licensing

• MS-DOS: Proprietary (commercial license owned by Microsoft).

• Windows: Proprietary closed-source software owned by Microsoft.

• Linux: Open-source (GNU GPL license), allowing anyone to view, modify, and distribute its code.

4. File Organization

• MS-DOS: Uses drive letters (e.g., C:\) with FAT file system and strict 8.3 filename conventions in legacy versions.

• Windows: Uses drive letters (C:\, D:\) primarily on NTFS file systems.

• Linux: Uses a unified single hierarchical directory tree rooted at '/' (e.g., /home, /etc, /usr) primarily with ext4 file systems.

5. Case Sensitivity

• MS-DOS: Case-insensitive (FILE.TXT and file.txt are identical).

• Windows: Case-preserving but generally case-insensitive by default.

• Linux: Strictly case-sensitive (File.txt, file.txt, and FILE.TXT are three distinct files).

6. Security

• MS-DOS: Minimal built-in security, user permissions, or memory protection.

• Windows: Includes User Access Control (UAC), BitLocker, Defender, and ACL permissions, but remains a frequent target for malware due to widespread usage.

• Linux: Highly secure architecture based on strict file permissions, root privilege access control, and rapid open-source vulnerability patching.

7. Common Uses

• MS-DOS: Legacy systems, embedded microcontrollers, and historical software/gaming execution.

• Windows: Personal computing, enterprise desktops, gaming, and business productivity software.

• Linux: Web servers, cloud infrastructure, supercomputers, Android mobile OS, embedded systems, and software development.